

Integrating paraffin phase change material in the storage tank of a solar water heater to maintain a consistent hot water output temperature

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ARTICLE INFO

Keywords:

Solar water heater
Encapsulated paraffin PCM
Steady water temperature maintenance
CFD simulation

ABSTRACT

The temperature of the hot water withdrawn for a solar water heater varies throughout a day. However, for process heat applications, constant temperature water can be required from a solar water heater. This is usually achieved by providing additional auxiliary heat input. Disadvantageously, such extra water heating incurs additional cost and, if met from fossil fuels, produces greenhouse gas emissions. An alternative approach of using a phase change material to moderate variations in the outlet temperature of hot water from the store is examined in this paper using an experimentally-validated CFD model of a solar water heater with a phase change material thermal energy storage in the hot water tank. The CFD model was solved by COMSOL Multiphysics. For a particular solar water heating system, incorporating an encapsulated paraffin wax phase change material has been shown to be able to deliver up to 1200 L of hot water at a temperature of $60 \pm 2 \text{ }^\circ\text{C}$ for more than 8 h.

Introduction

Background

For some process heat application, a water heater needs to provide water with a nearly constant temperature. As the temperature of the hot water withdrawn for a solar water heater can vary throughout a day, a consistent water temperature is usually achieved by providing additional auxiliary heat input. However, that extra water heating incurs additional capital and running costs and, if met from fossil fuels, produces greenhouse gas emissions. An alternative approach is the use of a phase change material (PCM) to moderate variations in the outlet temperature of hot water from the store. A schematic design of the system studied is shown in Fig. 1.

Solar radiation incident on each circular trough reflector is concentrated on the respective water-carrying absorber tube. The heated water passes to an encapsulated PCM-bed in a hot water thermal energy storage tank. When the temperature of the hot water produced (in periods of higher solar irradiation) reaches the melting temperature of the solid PCM, the PCM starts to melt. When the solar irradiation is lower (in

the afternoon) or is zero at night, the temperature of the outlet water from the collector is less than the PCM-bed temperature; the PCM-bed then transfers heat to the water maintaining a consistent water temperature at the PCM-bed outlet.

Objectives

The present work aims to determine if there are combinations of operational and material parameters that can enable an extended duration of the provision of hot water at a steady temperature within $\pm 2 \text{ }^\circ\text{C}$ fluctuation.

In the present work, the thermal performance of a storage tank filled with encapsulated paraffin PCM integrated with a solar water heater with circular-trough solar collectors is investigated. A comprehensive CFD simulation has been conducted at unsteady state conditions for encapsulated paraffin PCMs inside thermal energy storage (TES) tank in a solar water heating system. In a parametric study using the validated model, parameters were adjusted to enable hot water output with a nearly constant temperature can be achieved for extended periods.

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<https://doi.org/10.1016/j.seta.2021.101350>

Received 16 December 2020; Received in revised form 11 May 2021; Accepted 1 June 2021

Available online 10 June 2021

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Nomenclature

Alphabetical symbols Definition

C_p	specific heat capacity (J/kg.K)
d_p	particle diameter (m)
g	gravitational acceleration (m/s ²)
h	convective heat transfer coefficient (W/m ² .K)
I	initial turbulence intensity
k	thermal conductivity (W/m.K)
l	turbulence or eddy length scale
l	length (m)
m	weight of PCM in the bed (kg)
$\dot{m}$	mass flow rate (kg/s)
n	refractive index
N_t	number of total rays
Nu	Nusselt number
P	pressure (Pa)
P_k	production term
Pr	Prandtl number
q	initial ray position vector (m)
q	heat flux (W/m ²)
$\dot{Q}_b$	boundary heat source (W/m ²)
$\dot{Q}_{released}$	rate of heat released from PCM to the water in the bed (W)
q_{sf}	interstitial convective heat transfer coefficient (W/m ² .K)
$\dot{Q}_{stored}$	rate of heat stored by PCM in the bed (W)
r	ray position vector (m)
RE	relative error
Re_p	particle Reynolds number
r_p	particle radius (m)
T	temperature (K)
t	time (s)
u	velocity (m/s)
w	angular frequency (rad/s)

Subscripts

Subscript Definition

f	fluid
in	inlet
$loss$	thermal losses
out	outlet
pcm	phase change material
s	solid
w	water

Abbreviations

Abbreviation	Definition
CFD	Computational Fluid Dynamics
CPLS	Charging Period at Liquid State
CPPC	Charging Period at Phase Change
CPSS	Charging Period at Solid State
CTC	Circular Trough Collector
DPLS	Discharging Period at Liquid State
DPPC	Discharging Period at Phase Change
DPSS	Discharging Period at Solid State
PCM	Phase Change Material
RE	Relative Error
TES	Thermal Energy Storage

Greek Symbols

Greek Symbols	Definition
β	Bed pressure drop parameter
ε	Turbulent dissipation rate (J/kg.s)
ε	The emissivity of the absorber tube
ε_b	Bed porosity
μ	Dynamic viscosity (kg/m.s)
μ_T	Turbulent viscosity
θ	Phase fraction
κ	Porous medium permeability
k	Turbulent kinetic energy (J/kg)
ρ	Density (kg/m ³)

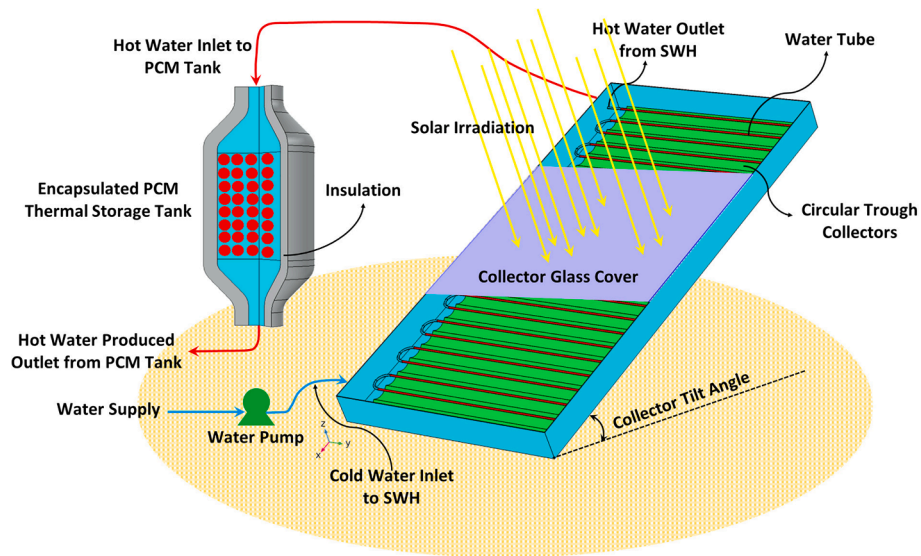


Fig. 1. The schematic diagram of the system examined.

Table 1
Thermophysical properties of paraffin [7].

Material Properties	Paraffin, Solid	Paraffin, Liquid
Melting temperature (°C)	60	
Latent heat of fusion (kJ/kg)	213	
Density (kg/m ³)	861	778
Specific heat (J/kg.°C)	1850	2384
Thermal conductivity (W/m.°C)	0.4	0.15

Relevant previous research

Phase change materials (PCMs), when compared to sensible heat materials have a higher energy storage density that is charged and discharged at the phase change temperature. Single tank packed-bed thermal energy storage systems incorporating PCMs have received significant attention from researchers [1,2]. Solid-liquid PCMs are more applicable than solid-gas and liquid-gas materials due to lower operating pressures [3]. Commonly-used solid-liquid PCMs can be organic, inorganic, eutectics, and metal materials. Criteria for selecting a suitable PCM include high latent heat, heat capacity, and thermal conductivity [4]. The advantages and disadvantages of salt hydrates, eutectic compounds, paraffin materials, and fatty acids as PCMs have been discussed extensively [5]. Paraffins are organic PCM materials that have several desirable thermal properties but also have low thermal conductivity and are flammable [6]. A paraffin TES system was designed and characterized by Nallusamy et al. [7]. Nitrate salts are the most common PCM materials that are used in high-temperature applications; however, most nitrate salts have large charging and discharging times due to their low thermal conductivity. Encapsulation of a PCM can enhance the heat transfer to the PCMs due to the increased heat transfer area. However, encapsulation of PCMs introduces pressure variations due to the phase change process in the capsule. Care must also be taken to ensure compatibility between the capsule covering and PCM materials to avoid corrosion [8].

In an experimental analysis of an integrating closed-loop thermosyphon solar water heater system with a PCM energy storage system, a PCM thermal storage system, the variation of water temperature was about 10 °C [9]. A cascaded energy storage system that combined PCM thermal with a water tank achieved temperatures in the range 47.5–57.5 °C [10]. Using different nanomaterial-embedded phase change materials and paraffin PCM as a TES in combination with a solar water heater system was found to improve the first and second laws efficiencies of the system [11]. A solar-to-thermal energy conversion system efficiency of 54.5% was found for a novel solar water heater combined with two kinds of composite PCM [12].

In a study of the performance of a solar water heater with a PCM storage tank in different climatic conditions, Ding et al. [13] found that the use of a PCM reduced both the power consumption of auxiliary heaters and heat losses. Xie et al. [14] concluded that for PCM in a tankless solar water heater, at a flow rate of 0.004 kg/s, the PCM could continue to heat water effectively after sunset. The addition of nanoparticles enhances the heat conduction rate of PCM [15], in addition, due to higher thermal conductivity, nanoparticles can decrease the charging/discharging time of PCM. Similarly expanded graphite has been used to create stearic acid/expanded graphite composite PCM with high thermal conductivity [16].

Using a 3D transient CFD model for a flat plate collector integrated with PCM, Badieli et al. [17] found that collector efficiency was enhanced from 33% to 46% with the PCM. Similar results have been found for an evacuated tube solar water heater integrated with PCM [18]. Manirathnam et al. [19] studied the thermal energy storage capacity of an all-glass evacuated tube solar water heater using a PCM and a nanocomposite PCM. They obtained that at 6:00 am the tank water temperature reached 45 °C without PCM, while with PCM and nanocomposite PCM, the tank water temperature increased up to 53 and

56 °C, respectively. The experimental performance of a system similar to that studied theoretically in this work has been examined by Madadi Avargani et al. [20].

Methodology

In the present work, a mathematical model for the PCM-bed was validated using experimental data reported by Nallusamy et al. [7]. After validation, the PCM-bed model was coupled to the overall solar water heating system model investigated previously by Madadi Avargani et al. [20].

The thermophysical properties for the paraffin PCM used in the simulation are listed in Table 1.

Mathematical model

Analysis

The coupled models of the solar water heating system and PCM-bed were solved simultaneously. The outlet temperature from the SWH unit at any time enters the PCM bed.

Details of the ray tracing modeling, governing equations, and assumptions used in the “Geometric Optics” ray tracing module in COMSOL to simulate solar radiation collection have been provided in previous work [20]. In summary, the optical parts of the solar collector are the glass cover, trough reflectors, and absorber tube. In optical modeling, the glass cover is a wall with angular dependant glass transmittance. Trough collectors are modeled as a surface with a specular reflectance. The absorber tube is modeled as a diffuse-scattering wall with a specified solar absorptance.

For a fixed water inlet rate, the variation of the fluid velocity inside the tube and also inside the bed due to the temperature variations during a period of the system operation was assumed to be negligible. To avoid excessive computation time, steady-state water velocity fields were used in coupling with heat transfer modeling.

In the fluid flow modeling, based on the Re number initially calculated for inlet properties of the water and tube dimensions, the water flow in the tube was then selected as either laminar or turbulent. For the system examined, a laminar flow was used for inlet water rates lower than 65 kg/h and turbulent flow selected for flow rates larger than 65 kg/h. The $k-\epsilon$ turbulent model to obtain the velocity field of water inside the tube is given according to the following equations [21]:

Continuity equation:

$$\rho \nabla \cdot \mathbf{u} = 0 \quad (1)$$

Momentum equation:

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot \left[-p\mathbf{I} + \left(\mu + \rho C_\mu \frac{k^2}{\epsilon} \right) (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \right] + \rho \mathbf{g} \quad (2)$$

In Eq. (2), C_μ is constant, k and ϵ are two dependent variables defined as turbulent kinetic energy and the turbulent dissipation rate, respectively. The transport equation for k and ϵ are given in Eqs. and respectively;

$$\rho(\mathbf{u} \cdot \nabla) k = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_k} \right) \nabla k \right] + \mu_T [\nabla \mathbf{u} : (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] - \rho \epsilon \quad (3)$$

$$\rho(\mathbf{u} \cdot \nabla) \epsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\epsilon} \right) \nabla \epsilon \right] + C_{\epsilon 1} \frac{\epsilon}{k} (\mu_T [\nabla \mathbf{u} : (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)]) - C_{\epsilon 2} \rho \frac{\epsilon^2}{k} \quad (4)$$

The model constants in Eqs. (2)–(4), C_μ , $C_{\epsilon 1}$, $C_{\epsilon 2}$, σ_k , and σ_ϵ are valued as 0.09, 1.44, 1.92, 1.0, and 1.3 respectively [21].

The continuity and momentum equations for the air layer between the absorbers and glazed aperture cover in the solar energy collector are given in Eqs. and respectively:

Table 2

The optical properties and heat transfer input parameters for CFD simulation.

Parameter	Value	Parameter	Value/Range
Geometrical Optics		Heat Transfer	
Number of sun rays	100,000	Absorber tube	Copper
Refractive index of exterior domains	1	The mass flow rate of water	0.005–0.15 kg/s
Maximum number of secondary rays	0	Inlet temperature	288–303 K
Collector surface slope error	2 mrad	The emissivity of the absorber tube	0.1
Maximum disc angle	4.65 mrad [26]	Wind velocity	0–1.5 m/s
Solar irradiation intensity	1000 W/m ²	Ambient temperature	293–303 K
Glass transmittance	0.92	PCM packed-bed diameter	0.2–0.4 m
The absorption coefficient of the tube (Black copper)	0.9	PCM packed-bed height	0.2–1 m
Collector reflectance coefficient	0.85	PCM packed-bed porosity	0.2–0.5
		PCM capsule size (diameter)	55 mm

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0 \quad (5)$$

$$\rho \frac{\partial u}{\partial t} + \rho (\nabla \cdot u) u = \nabla \cdot \left[-pI + \mu (\nabla u + (\nabla u)^T) - \frac{2}{3} \mu (\nabla \cdot u) I \right] + \rho g \quad (6)$$

The water flow in the porous medium in PCM packed bed is governed by the continuity and momentum equations;

$$\rho \nabla \cdot u = 0 \quad (7)$$

$$0 = \nabla \cdot \left[-pI + \mu \frac{1}{\varepsilon_b} (\nabla u + (\nabla u)^T) - \frac{2}{3} \mu \frac{1}{\varepsilon_b} (\nabla \cdot u) I \right] - (\mu \kappa^{-1} + \beta \varepsilon_b \rho |u|) u + \rho g \quad (8)$$

where μ is the dynamic viscosity of the water, ε_b is the bed porosity, ρ is the density of the water, κ is the permeability of the porous medium, and β is the bed pressure drop parameter and is defined as:

$$\beta = \frac{c_F}{\sqrt{\kappa}} \quad (9)$$

In Eq. (9), c_F is the dimensionless Forchheimer parameter.

For a bed porosity ε_b and average particle diameter d_p , the permeability of the porous medium is given by [22];

$$\kappa = \frac{d_p^2}{180} \frac{\varepsilon_b^3}{(1 - \varepsilon_b)^2} \quad (10)$$

In Eq. (8), u is calculated from [23];

$$u = - \frac{d_p^2}{180 \mu} \frac{\varepsilon_b^3}{(1 - \varepsilon_b)^2} (\nabla p) \quad (11)$$

The general unsteady-state governing equation for heat transfer modeling in solid domains is;

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p u \cdot \nabla T + \nabla \cdot (-k \nabla T) = 0 \quad (12)$$

where ρ is density, C_p is the heat capacity at constant pressure, and k is the thermal conductivity.

For the porous medium PCM packed-bed, the corresponding heat transfer equations are [24];

(i) encapsulated PCM thermal energy storage tank;

$$(1 - \varepsilon_b) \rho_s C_{p,s} \frac{\partial T_s}{\partial t} = \nabla \cdot ((1 - \varepsilon_b) k_s \nabla T) + q_{sf} (T_f - T_s) \quad (13)$$

(ii) water through PCM thermal energy storage tank;

$$\varepsilon_b \rho_f C_{p,f} \frac{\partial T_f}{\partial t} + \varepsilon_b \rho_f C_{p,f} u_f \cdot \nabla T_f = \nabla \cdot (\varepsilon_b k_f \nabla T) + q_{sf} (T_s - T_f) \quad (14)$$

In Eqs. (13) and (14), u_f is the water velocity field in the porous medium that is calculated from Eq. (11), the last terms on the right side are the heat source due to the heat transferred from the paraffin-filled capsules to the water and q_{sf} is the interstitial convective heat transfer coefficient. The q_{sf} for spherical capsules can be estimated as:

$$q_{sf} = \frac{6(1 - \varepsilon_b)}{d_p} h_{sf} \quad (15)$$

In Eq. (15), the interstitial heat transfer coefficient h_{sf} is calculated from the correlation;

$$\frac{1}{h_{sf}} = \frac{2r_p}{k_f Nu} + \frac{2r_p}{\beta k_s} \quad (16)$$

For spherical capsules, $\beta = 10$, and Nu is the Nusselt number derived from the correlation [25]:

$$Nu = 2 + 1.1 Pr^{1/3} Re_p^{0.6} \quad (17)$$

where Prandtl number Pr and particle Reynolds number Re_p are defined by:

$$Pr = \frac{\mu C_{p,f}}{k_f} \quad (18)$$

$$Re_p = \frac{2r_p \rho_f |u_f|}{\mu} \quad (19)$$

In the phase change process, the properties of the solid and liquid states of the PCM are used in governing equations of the porous medium TES, Eqs. (13) and (14), based on the following equations:

$$\rho = \frac{\rho_{PCM,l} + \rho_{PCM,s}}{2} \quad (20)$$

$$C_p = \theta_l C_{p,PCM,l} + \theta_s C_{p,PCM,s} + \Delta H_{s-l} \frac{\partial \alpha_m}{\partial T} \quad (21)$$

$$\alpha_m = \frac{\theta_s - \theta_l}{2(\theta_s + \theta_l)} \quad (22)$$

$$k = \theta_s k_s + \theta_l k_l \quad (23)$$

$$\theta_s + \theta_l = 1 \quad (24)$$

where θ_l and θ_s are the fractions of the PCM in liquid and solid states, respectively, and ΔH_{s-l} is the latent heat of the phase change.

Boundary conditions

At the inlet of the solar collector, the mass flow rate and constant inlet temperature were used.

$$- \int \rho(u \cdot n) dS \Big|_{inlet, SWH} = \dot{m} \quad (25)$$

$$T|_{inlet, SWH} = T_{in} \quad (26)$$

At the inlet of the PCM thermal energy storage tank, the water outlet from the solar collector enters the PCM bed, therefore,

$$- \int \rho(u \cdot n) dS \Big|_{inlet, PCM} = \dot{m} \quad (27)$$

$$T|_{inlet, PCM} = T|_{outlet, SWH} \quad (28)$$

At the absorber tube surface, the collected heat on the external

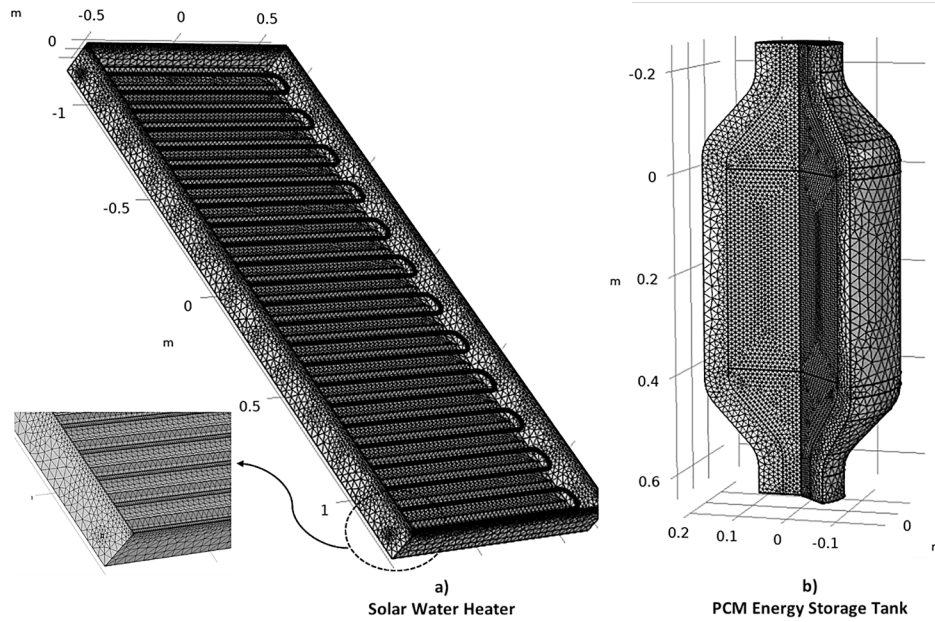


Fig. 2. Generated meshes for the CFD simulation of the solar collector and PCM.

Table 3

Results of mesh independence analysis for PCM packed-bed system.

Test Samples	Domain elements	Boundary elements	Relative error, %
1	2560	273	4.23
2	3864	341	3.75
3	6499	443	2.35
4	16,618	774	2.72

surface of the absorber tube is used that is estimated from Eq. (29):

$$-n \cdot k \nabla T|_{\text{absorber pipe surface}} = Q_b \quad (29)$$

At the outlet of the water tube in the solar collector and PCM bed, the constant zero pressure is specified.

Simulation parameters

The optical properties of circular trough collectors (CTCs), water tube, and glass cover together with input parameters for heat transfer modeling are given in Table 2. The outer wall of the PCM thermal energy storage tank is insulated by a 5 cm thick glass wool. The initial temperature of the system is the same as the ambient temperature at the start of the simulation.

Inlet water rate, bed porosity, and system geometrical dimensions affect system performance. So the main parameters investigated are the inlet water rate and bed length. Simulations were conducted for solar irradiation data for the 15th July 2019 in Isfahan, Iran at the latitude and longitude of 32.66 and 51.68 respectively.

Mesh independency study and model validation

The generated meshes for the solar collector and PCM-bed tank modeling are shown in Fig. 2 a and b respectively. A mesh independence study and model validation of solar collector has been reported in detail in previous work [20].

In CFD modeling of PCM energy storage tank, a mesh controlling parameter was defined for free tetrahedral geometry mesh generation. To study the mesh independence study of PCM, the outer walls of the bed were considered ideally insulated. A constant inlet rate of water with a fixed temperature was assumed to enter the PCM-bed with specified initial temperature and specified operating time.

The independence criterion is defined as the summation of the heat gained by the water trapped between the PCM capsules and PCM solid in the period t must be equal to the net heat lost by water stream through the PCM with less than 3% relative error. The relative error is defined as:

$$RE = \frac{\dot{m}_w C_{p,w} (T_{in,PCM} - T_{out,PCM}) - (\dot{m}_{w,bed} C_{p,w} + \dot{m}_{PCM,bed} C_{p,PCM}) [T_{ave,bed,t} - T_0] / t}{\dot{m}_w C_{p,w} (T_{in,PCM} - T_{out,PCM})} \quad (32)$$

To evaluate the performance the following storage rates are defined; When charging, the rate of energy stored by the PCM tank is calculated from;

$$\dot{Q}_{\text{stored}} = \dot{m}_w C_{p,w} T_{w,in,PCM} - \dot{m}_w C_{p,w} T_{w,out,PCM} - \dot{Q}_{\text{loss,PCM}} \quad (30)$$

When discharging, the rate of energy released from the PCM to the water is calculated from;

$$\dot{Q}_{\text{released}} = \dot{m}_w C_{p,w} T_{w,out,PCM} - \dot{m}_w C_{p,w} T_{w,in,PCM} + \dot{Q}_{\text{loss,PCM}} \quad (31)$$

From the results of mesh independence analysis for the PCM tank given in Table 3., the 3rd mesh elements sample is selected for final simulations. The PCM storage tank model was compared with experimental data obtained in [7]. The results of this model validation shown in Fig. 3, together with standard error, show that the model can predict water and PCM-bed temperatures with acceptable accuracy.

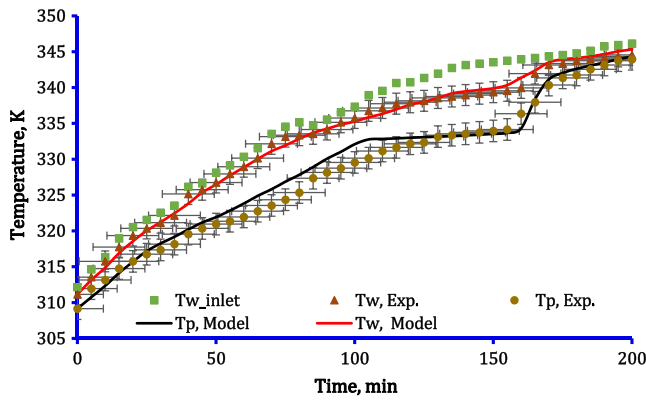


Fig. 3. PCM packed-bed model validation with standard error.

Results

Temperature variation in PCM

The variations of water and PCM temperatures at the midpoint and outlet of the bed from 7:00 to 22:00 are shown in Fig. 4. As the solar irradiation increases from 7:00 to 13:00, the temperature of the outlet water from the collector increases gradually. After 13:00 the temperature of the outlet water from the collector decreases until sunset due to less solar irradiation. It continues to reduce at night due to lower ambient temperatures. The period required to warm up the PCM-bed depends on the system conditions and parameters. For the conditions encountered, this period is about one hour for PCM at the bottom of the

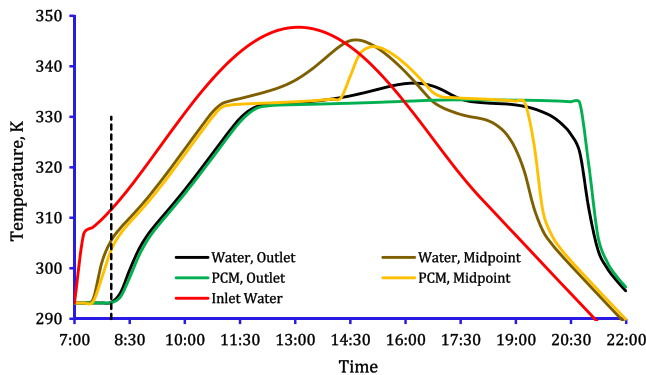


Fig. 4. The variations of the PCM and water temperature inside the PCM packed-bed, $\dot{m}=35$ kg/h, $\varepsilon_b=0.49$, $L_b=0.6$ m, $r_b=0.15$ m, $T_{in,SWH}=293$ K, and $T_0=293$ K.

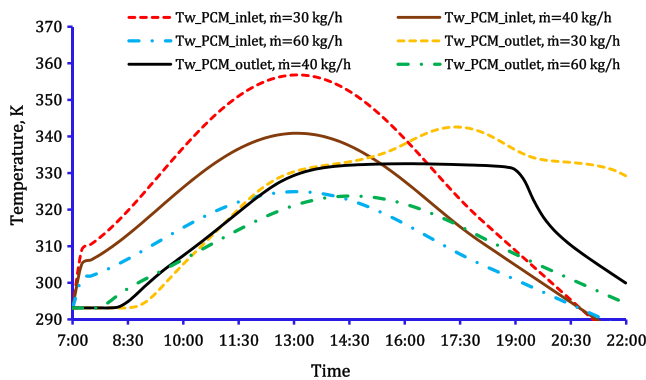


Fig. 5. The effect of the inlet water rate on the temperature of hot water produced outlet from SWH and PCM units, $\varepsilon_b=0.49$, $L_b=0.6$ m, $r_b=0.15$ m, $T_{in,SWH}=293$ K, and $T_0=293$ K.

bed as shown in Fig. 4. The temperatures of the water passing through the porous medium and PCM increased gradually due to an increase in the inlet temperature from sunrise up to midday until the PCM temperature reaches its melting temperature of 60°C . It then remains nearly constant during the PCM phase change process. After that, when the whole PCM has melted in each section of the bed, the then liquid PCM gains sensible heat. When the water temperature inside the PCM bed drops due to the falling inlet temperature in the afternoon, some of the stored energy in the PCMs is released to the water inside the bed. This leads to the stability of the outlet water temperature from the PCM bed.

Effect of inlet water flow rate

The effects of the inlet water rate on the outlet temperature of hot water produced from the solar collector and PCM units are shown in Fig. 5. The results show that it is possible to obtain hot water with a nearly constant outlet temperature over a long period at specific low input mass flow rates. Large inlet water rates are not appropriate due to the large changes in water temperature at the PCM bed outlet. It can be inferred that there is a specified and limited range of inlet water rate to the system that the water temperature at the PCM outlet can be constant over an extended period of system operation.

Fig. 5 also indicates that the water flow rate influences the phase-change transition process. When higher inlet water rates enter the solar collector, the outlet water temperature is lower. Therefore, for larger rates, the PCM experiences a lower temperature and the charging process starts at an earlier time, and the time required for the complete charging process is greater. For example, for 30 kg/h, the charging process of the PCM at the bottom of the bed starts at about 9:00, while for 60 kg/h it starts at about 8:00.

The fraction of the liquid phase in the PCM storage tank for various times and water rates of 30 and 40 kg/h are shown in Fig. 6. By comparing the results in Fig. 5 and Fig. 6, it can be observed that, for example, for a low flow rate of 30 kg/h and at 17:30, the PCM bed is liquified completely so the average temperature of the liquid PCM in the bed may be higher than its melting point. In contrast, for higher flow rates, the entire PCM in the bed may not reach its melting point at any time during system operation. For large flow rates, the water at a temperature near the PCM melting point may not be produced by the solar collector. Therefore, to achieve hot water at the outlet of the PCM bed with a constant temperature for a considerable period of the day, a specified inlet rate must be selected.

The PCM bed temperature profiles corresponding to the liquid phase fraction in the PCM bed shown in Fig. 6, are illustrated in Fig. 7 for inlet rates of 30 and 40 kg/h. By comparing the results, it clearly can be seen that for 30 kg/h, the average temperature of the PCM bed is greater than its melting point at about 17:30, and the entire bed is liquified at this time. On the other hand, by focusing on the results obtained in Fig. 7, it can be found that the water temperature at the PCM outlet for 30 kg/h is not constant at a long period of system operation. While, for 40 kg/h, the outlet water temperature is almost constant from about 12:00 to 19:30. In these conditions, about 300 L of water with a temperature of 60°C can be produced only by a single solar collector.

The temperature contours of the solar collector and PCM units for specified operating conditions and at the various operating times are shown in Fig. 8. It can be seen that for inlet water rate of 40 kg/h, the temperature of the hot water produced and outlet from the PCM tank is approximately constant from 13:00 up to 18:00.

Six periods of PCM behavior can be identified;

- When the solid material absorbs thermal energy to reach its melting point, this period is the charging period at solid-state (CPSS).
- After that, the solid material absorbs energy at the melting point to convert it to the liquid state. This period is the charging period at the phase change process (CPPC).

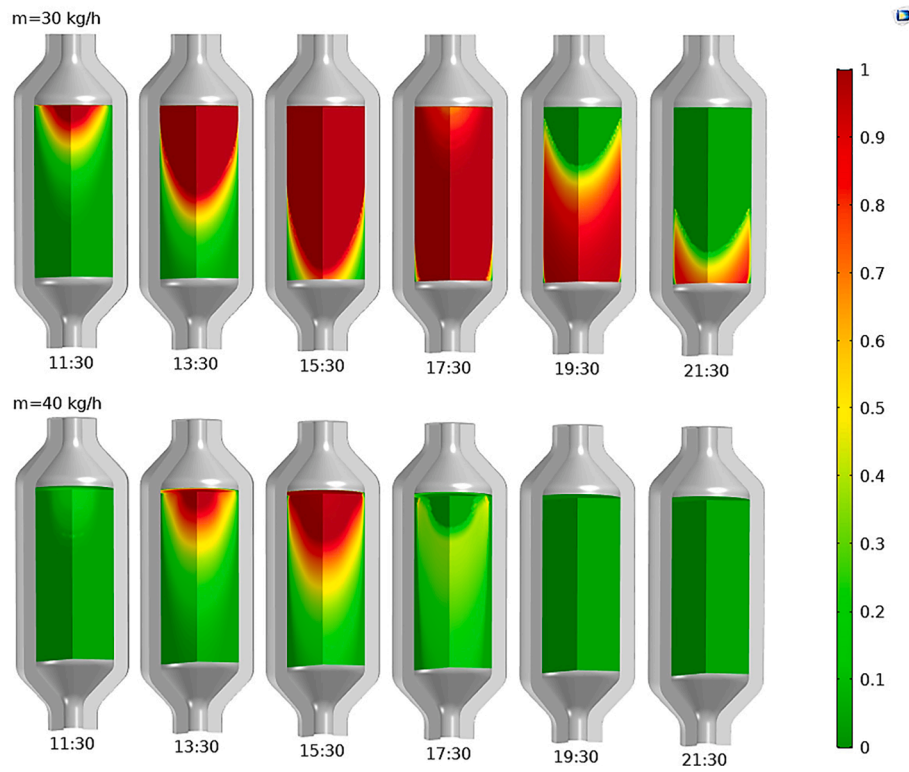


Fig. 6. The liquid phase fraction in the PCM packed-bed at various times during a day, and for rates of 30 and 40 kg/h, $\varepsilon_b=0.49$, $L_b = 0.6$ m, $r_b = 0.15$ m, $T_{in,SWH} = 293$ K, and $T_0 = 293$ K.

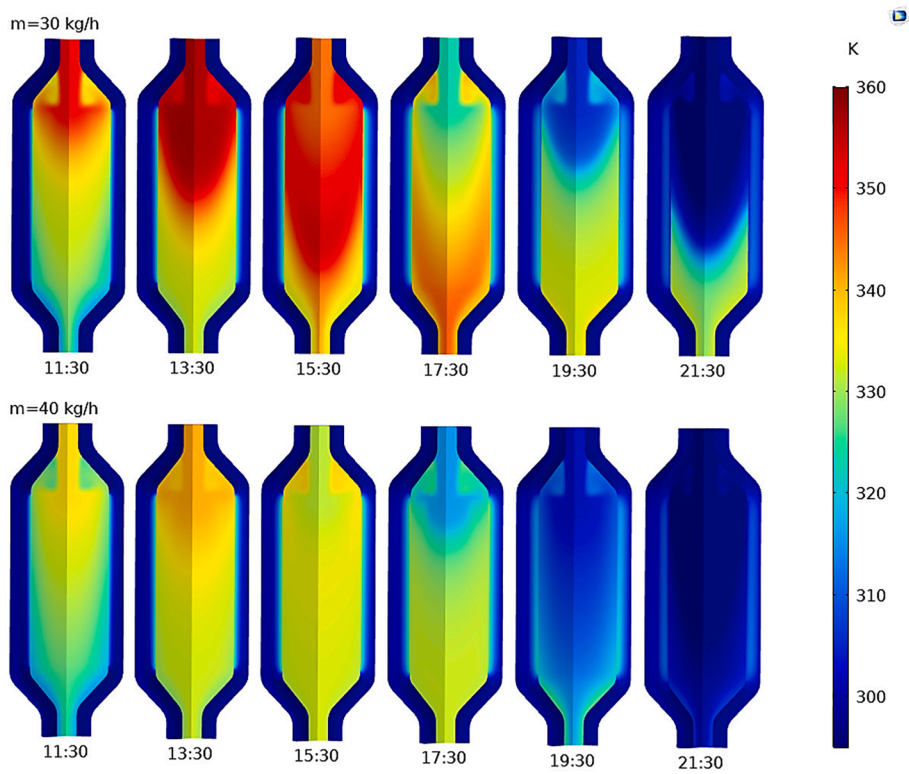


Fig. 7. The temperature contours of PCM packed-bed at various times during a day, and for water mass flow rate of 30 and 40 kg/h, $\varepsilon_b=0.49$, $L_b = 0.6$ m, $r_b = 0.15$ m, $T_{in,SWH} = 293$ K, and $T_0 = 293$ K.

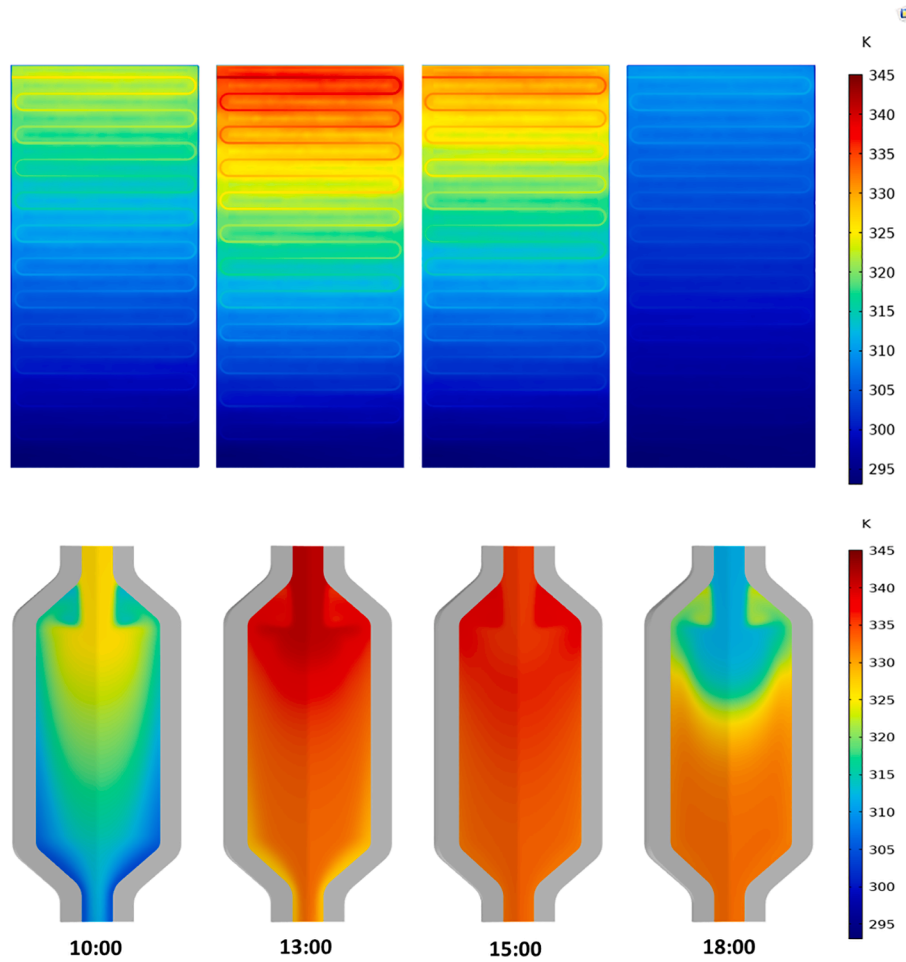


Fig. 8. The temperature profiles of the solar collector and PCM units at various times during a day, and for $\dot{m}=40 \text{ kg/h}$, $\epsilon_b = 0.49$, $L_b = 0.6 \text{ m}$, $r_b = 0.15 \text{ m}$, $T_{\text{in.SWH}} = 293 \text{ K}$, and $T_0 = 293 \text{ K}$.

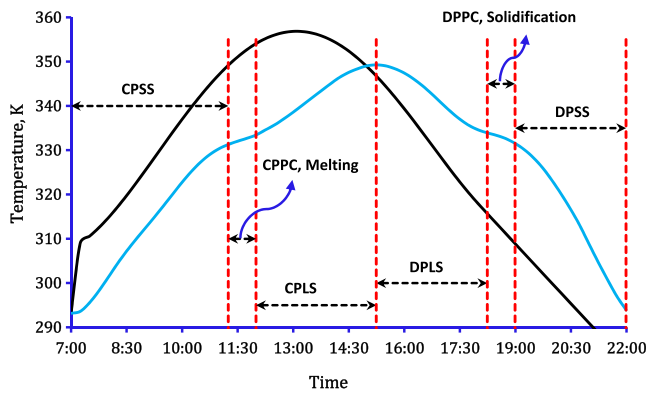


Fig. 9. Charging/discharging periods in solid and liquid states and the phase change processes for inlet water rate of 30 kg/h , $L_b = 0.6 \text{ m}$, $\epsilon_b = 0.49$, $r_b = 0.15 \text{ m}$, $T_{\text{in.SWH}} = 293 \text{ K}$, and $T_0 = 293 \text{ K}$.

- If the liquefied PCM absorbs energy to reach a temperature higher than the melting point, this is the charging period at the liquid state (CPLS).
- When a liquefied PCM at a temperature higher than the melting point releases its thermal energy to reach the melting point, this period can be defined as the discharging period at liquid state (DPLS).

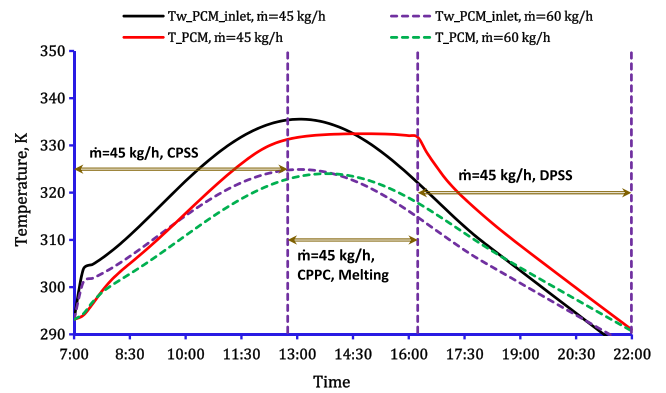


Fig. 10. Charging/discharging periods in solid and liquid states and the phase change processes for different inlet water rates, $L_b = 0.6 \text{ m}$, $\epsilon_b = 0.49$, $r_b = 0.15 \text{ m}$, $T_{\text{in.SWH}} = 293 \text{ K}$, and $T_0 = 293 \text{ K}$.

- A liquefied PCM at its melting point releases its thermal energy to be solidified, this period is the discharging period in the phase change processes (DPPC).
- After that, the removal of PCM thermal energy occurs in the solid state. This period can be the discharging period in the solid-state (DPSS).

These six periods may all occur for a particular set of operational and

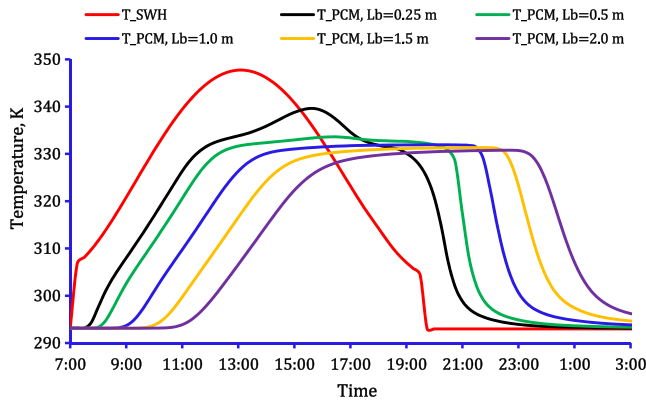


Fig. 11. The effect of PCM bed length on the hot water temperature outlet from the solar collector and PCM units, $\dot{m}=35$ kg/h, $\varepsilon_b=0.3$, $r_b=0.15$ m, $T_{in, SWH}=293$ K, and $T_0=293$ K.

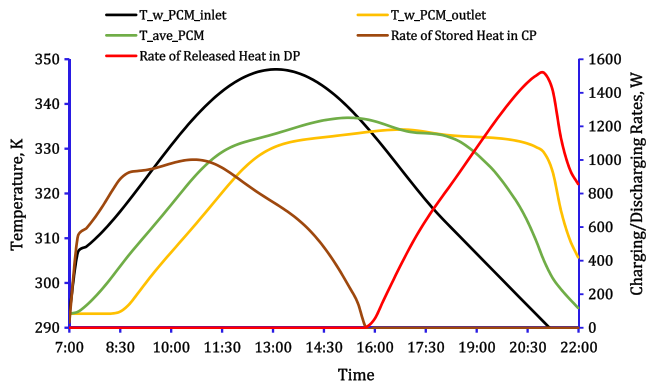


Fig. 12. The rates of charging/discharging processes and variation of outlet temperature from the SWH and PCM units, and the average temperature of PCM in the bed during a day, $\dot{m}=35$ kg/h, $\varepsilon_b=0.4$, $r_b=0.15$ m, $L_b=0.6$ m, $T_{in, SWH}=293$ K, and $T_0=293$ K.

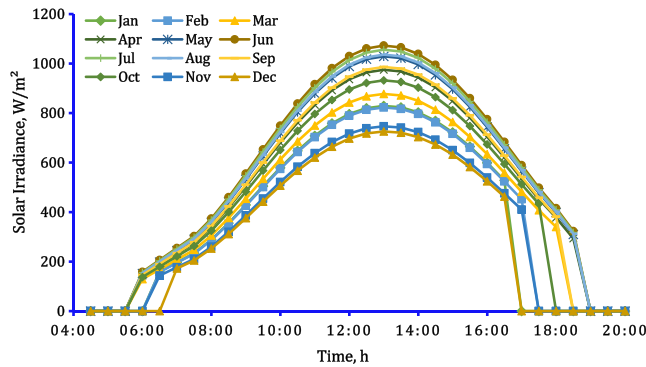


Fig. 13. The normalized solar irradiance for the 15th day of different months of 2019, for Isfahan city, Iran.

material parameters, or only some of them may occur under other conditions. These six charging/discharging periods for the present system for different inlet water rates are shown in Fig. 9 and Fig. 10.

The results in Fig. 9 indicate that for smaller inlet rates, all six periods for thermal energy exchange between PCM and water in the bed occur and therefore, hot water with a constant temperature is not produced for a long period. For larger inlet water rates, as shown results in Fig. 10, in some cases, only some of these periods occur. For example, for the inlet rate of 45 kg/h, only three periods of CPSS, CPPC, and DPSS occur. For

larger flow rates, since the outlet water from the solar collector did not reach the melting point of solid PCM, therefore only charging/discharging processes in the solid-state occur in form of sensible heat. In these conditions, the hot water with a constant temperature cannot be produced at any period of system operation. To achieve the best performance of the present system, parameters must be adjusted in such a way that the system operates for a long duration in CPPC and DPCC periods.

Effect of PCM-bed length

The effect of PCM-bed length on the hot water produced by the system and charging/discharging periods is shown in Fig. 11. For smaller PCM-bed lengths, the charging process starts earlier so the difference between hot water temperature at the solar collector and PCM outlets is smaller compared to larger lengths. Also, for smaller lengths, as phase change and discharging periods start earlier, temperature stabilization for smaller lengths is more transient than larger lengths. For small lengths, the variations of the water temperature at the outlet of PCM are also more than larger lengths, and produced hot water has a constant temperature for a short period.

For larger lengths, the charging process starts later so the temperature difference between hot water produced by solar collector and outlet from the PCM-bed is larger. For bed length of 0.5 m, the hot water produced has a constant temperature of 60 °C from about 12:30 to about 20:30, and in this period about 280 L hot water can be produced. For bed length of 1 m, hot water with a constant temperature is produced from about 14:45 to about 21:45. For this amount of PCM, about 245 L hot water can be produced. For PCM-bed length of 2 m, the constant period of hot water produced starts from about 17:00 and takes along up to about 23:30. For these conditions about 227.5 L hot water is produced.

By selecting a proper bed length, the time of hot water required can be managed based on demands. Larger amounts of PCM in the bed do not necessarily produce more hot water but can shift the time of hot water produced with a constant temperature during the day to midnight. By adjusting the amount of PCM in the bed, that the period in which hot water is needed during the day and night can be managed.

Rates of heat charging and discharging

In the charging process, a part of the thermal energy of water heated by the solar collector is transferred to the PCMs and a fraction of inlet water energy is stored in the PCMs. On the other hand, when the temperature of the water outlet from the SWH unit decreases due to less solar irradiation in the afternoon, and the average temperature of the water in the PCM bed or at the outlet of the bed is greater than inlet temperature, water passing through the bed absorbs some of the heat energy from the PCMs to prevent a temperature drop at the outlet of the PCM bed. The rates of heat exchange between the PCMs and water passing through the bed in charging/discharging processes are shown in Fig. 12.

The values of charging/discharging rates are shown on the right side of Fig. 12, and the temperature amounts of the water at the inlet and outlet of the PCM bed are shown on the left side of Fig. 12. When the temperature of the water outlet from the solar collector and inlet to the PCM units drops due to a reduction in solar energy, and water inlet temperature is lower than the outlet temperature from the PCM bed (the intersection of two yellow and black lines), the charging process ends, and the discharging process starts. At the start of the discharging process, the rate of thermal energy released from the PCMs to the water passing through the bed is small and after that, with a sharp reduction in solar irradiation in the afternoon and a consequently significant reduction in the water inlet temperature to the PCM bed, the rate of heat release in the discharging process increases until reaching its maximum value at the time that solar irradiation is zero. For these conditions, it can be seen that the temperature of the water entering the bed at about

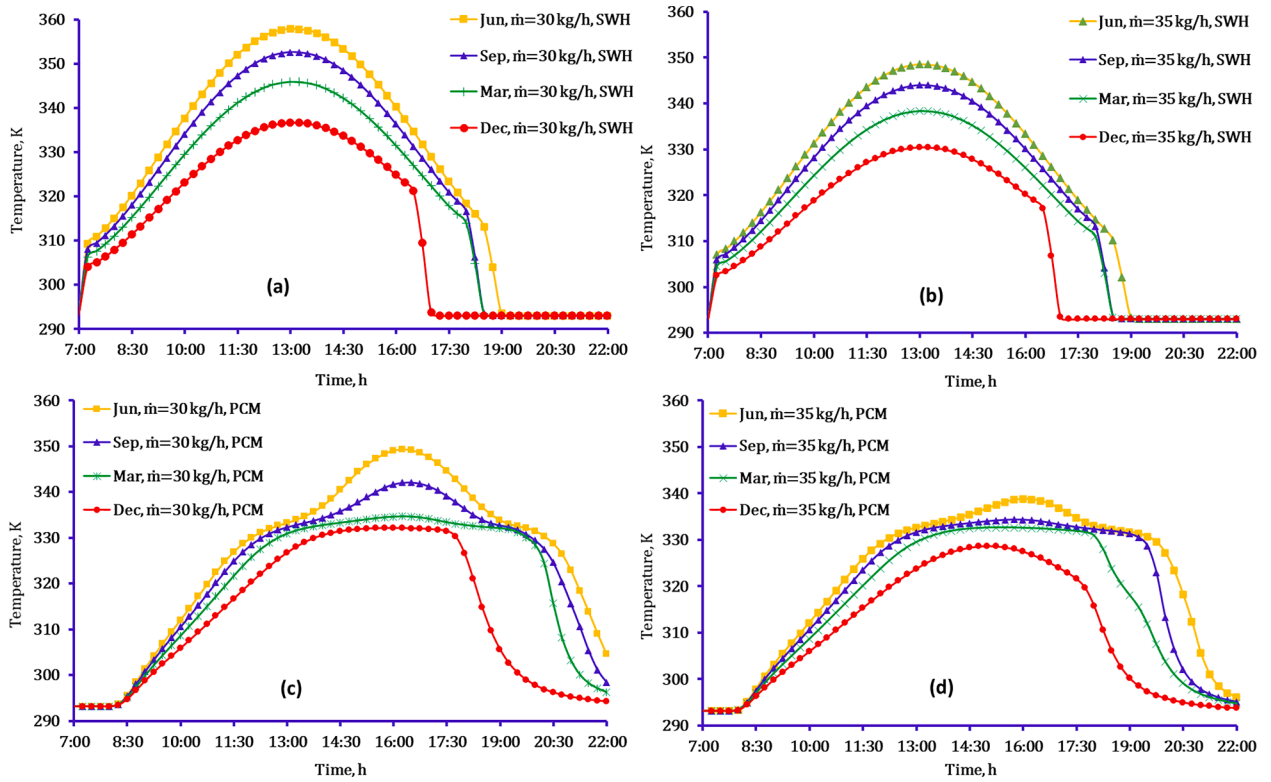


Fig. 14. The outlet temperature of hot water produced from SWH and PCM units for December, March, September, and June, $\varepsilon_b = 0.49$, $r_b = 0.15$ m, $L_b = 0.6$ m, $T_{in, SWH} = 293$ K, and $T_0 = 293$ K.

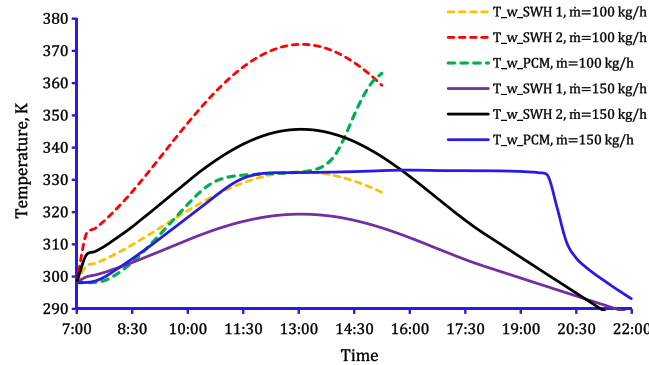


Fig. 15. The effect of the inlet water rate for two series multiple SWH units on the temperature of hot water produced outlet from SWHs and PCM bed, $\varepsilon_b = 0.5$, $L_b = 1.0$ m, $r_b = 0.2$ m, $T_{in, SWH 1} = 293$ K, and $T_0 = 293$ K.

15:00 is lower than the temperature of the water leaving the bed; at this moment the charging process ends and the discharging process begins. These results show that for these conditions hot water with a constant temperature can be produced for an extended period. The discharging process plays a key role in temperature stabilization. By decreasing the inlet water rates, the end time of the charging process, and the starting time of the discharging process shift to later times, and the charging process takes more time and discharging process starts later. The results in Fig. 12 also show that the maximum rate of heat energy stored in the PCM is less than 1000 W, while at 21:00 the rate of heat energy released from the PCM into the water is more than 1500 W. This means that for the selected parameters and conditions, the discharge process plays its role well, and although the maximum rate of heat gained by solar energy during the operation of the system is less than 1050 W, in the discharge process sometimes up to 1500 W of heat energy can be transferred from PCM to water to prevent a temperature decrease.

Annual system performance with a single solar collector

To investigate the capability of the system at different times of operation, the performance of the system is discussed in the last month of each season. Solar irradiation for the 15th of each month (data in Fig. 13) is used in the model simulation.

The temperature profiles for outlet water from solar collector and PCM units for December, March, September, and June, and for inlet water rates of 30 and 35 kg/h are shown in Fig. 14. For a water rate of 30 kg/h and for a single collector, the water outlet temperature from the SWH system in June can reach up to 346 K, while at the same conditions, the temperature of the water at the outlet of the SWH unit in September reaches a maximum of 337 K at midday time. In the months of the year with smaller solar irradiation, the production of hot water with a nearly constant temperature is possible with the combined system, when the operational parameters are correctly adjusted. For example, although

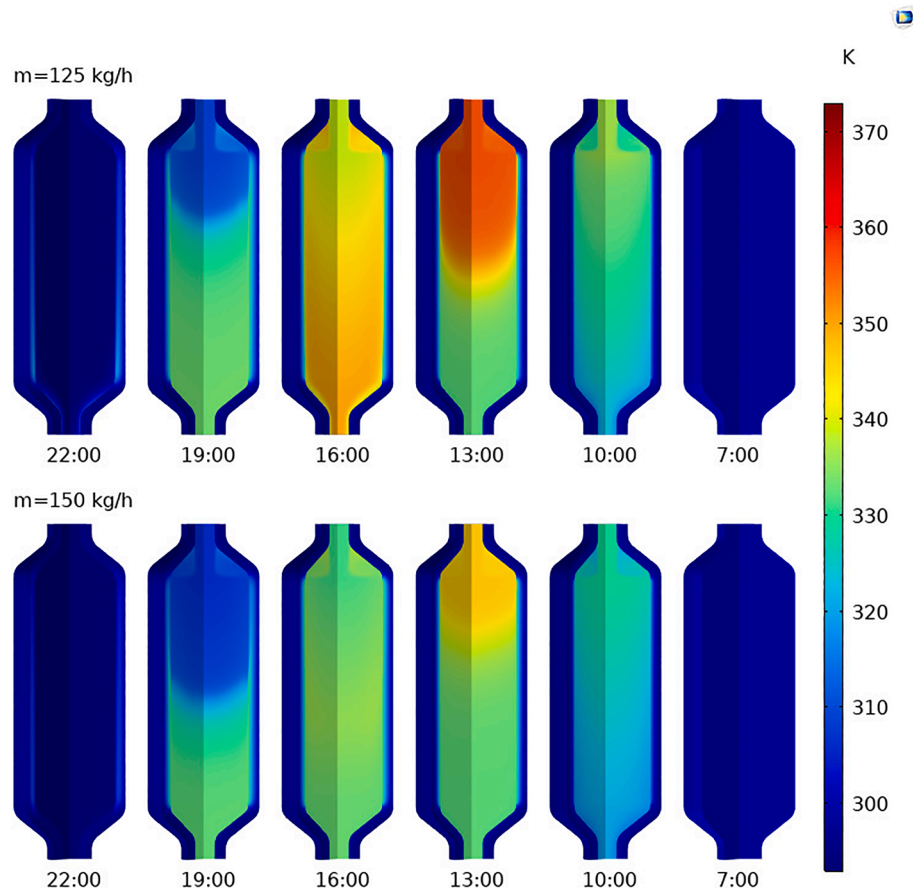


Fig. 16. The temperature profiles of PCM bed for two consecutive SWH units, $\varepsilon_b = 0.49$, $L_b = 1.0$ m, $r_b = 0.2$ m, $T_{in,SWH\ 1} = 293$ K, and $T_0 = 293$ K.

with a rate of 35 kg/h a period with constant temperature cannot be obtained in December, with 30 kg/h greater than 4 hrs hot water with a nearly constant temperature can be produced. This is while with this amount of inlet rate, hot water with constant temperature cannot be produced in June. In June with a rate of 43 kg/h, in September with a rate of 35 kg/h, and in March with a rate of 32 kg/h about 6 hrs hot water with a nearly constant temperature close to the PCM melting temperature with a single solar collector can be produced.

System performance with two solar collectors in series

To meet large hot water loads, multiple solar collectors are used in series. Outlet water temperature variations from two solar collectors in series and temperatures at the outlet from the PCM-bed for an inlet mass flow rate of 100 and 150 kg/h are shown in Fig. 15.

From the results in Fig. 15, for an inlet water rate of 100 kg/h, the maximum outlet temperature of the first solar collector can reach 335 K, while at the outlet of the second solar collector, the water reaches boiling point. For smaller inlet water rates, similar to a single solar collector, the temperature variation of the outlet water from the PCM bed is large, with no desired conditions for hot water production achieved.

With the correct selection of inlet water rate, for example, 150 kg/h, an extended duration from about 11:45 up to 19:45, hot water is produced by the system with a temperature of $60\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$. In this period about 1200 L hot water can be produced, while for a single collector, the maximum amount of hot water produced was less than 400 lit a day for the same conditions.

The temperature profiles of the PCM bed for two solar collectors in series and inlet water rate of 125 and 150 kg/h are shown in Fig. 16. For smaller inlet rates the temperature of the PCM bed can reach

temperatures higher than the melting point of PCM. In these conditions, the liquified PCM in the bed can absorb more thermal energy in sensible form so its temperature increases more than the melting point. In these cases, a sensible heat charging process in the liquid state occurs, with consequently, a discharging process also in the liquid state. As a result, the temperature of the water inside the bed and in contact with PCM changes during system operating, and the system cannot produce hot water with a constant temperature for an extended period.

Comparing a single collector, in Fig. 5, with two solar collectors in series, in Fig. 15; for two solar collectors, more than twice the volume of water can be heated to the desired temperature for a longer period.

Temperature profiles and liquid phase fractions of the PCM in the PCM-bed heated by two collectors in series are shown in Figs. 16 and 17 respectively. These results show that for an inlet water rate of 125 kg/h, the PCM is completely liquified at 16:00. At greater inlet rates, for example, 150 kg/h, conditions do not enable the entire PCM to liquefy at any time of operation.

Conclusions

With appropriate parameter selection, integration of a PCM-bed in the storage tank of a solar water heating system unit, hot water can be produced with a consistent temperature near the melting temperature of the PCM for an extended period of system operation.

For a single solar collector integrated with a PCM-bed in the hot water store, for 7 h of system operation, with bed diameter of 0.3 m, and length of 0.6 m, up to 300 L hot water with a temperature of $60\text{ }^{\circ}\text{C}$ and $\pm 2\text{ }^{\circ}\text{C}$ can be produced. The results of the annual system performance investigation showed that hot water production with a constant temperature throughout the year can be achieved with an appropriate selection of effective parameters.

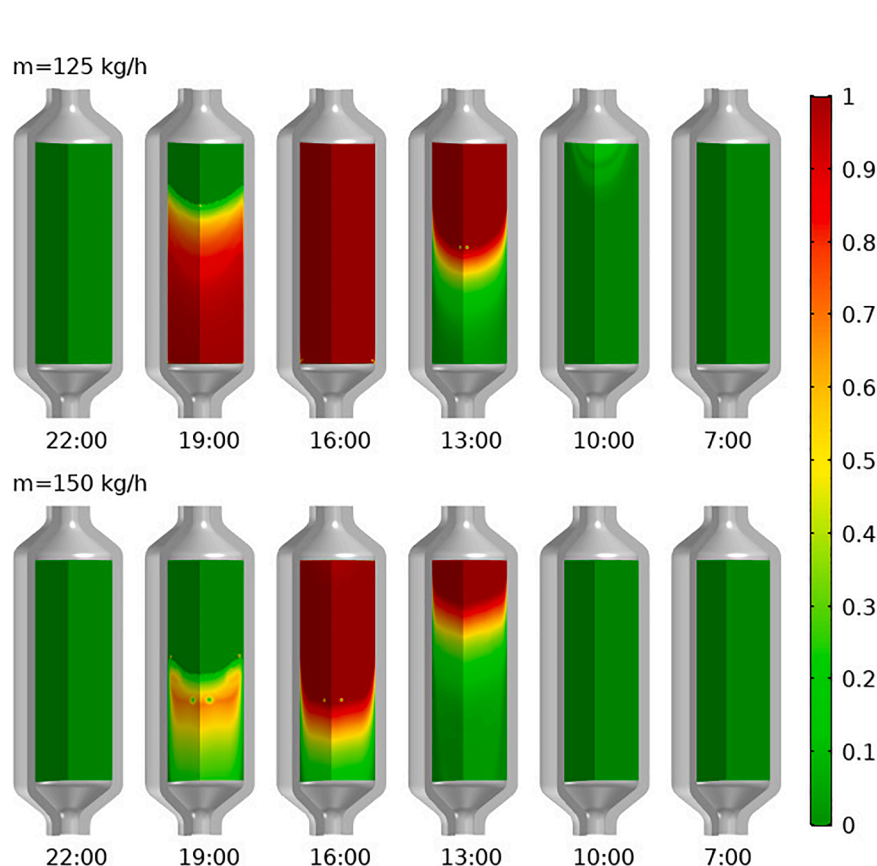


Fig. 17. The fractions of the liquid phase of PCM in the bed for two consecutive SWH units, $\varepsilon_b=0.49$, $L_b = 1.0$ m, $r_b = 0.2$ m, $T_{in,SWH 1} = 293$ K, and $T_0 = 293$ K.

At high flow rates of water through the system, the hot water produced at the outlet of the bed does not reach the melting temperature of PCM as a result, hot water cannot be produced at a constant temperature. To overcome this, multiple collectors in a series configuration may be used. For two solar collectors in series, eight hours of the system operation producing 60 °C hot water can be achieved. By adjusting parameters, hot water production time can be shifted from midday to midnight. For these conditions up to 1200 L of hot water can be produced. By using two consecutive collectors, the performance of the system is increased far more than two-fold over that of a single solar collector. The selection of the number of collectors in the series configuration depends significantly on the amount of hot water required and the latent heat and melting point of the phase change material.

CRediT authorship contribution statement

Vahid Madadi Avargani: Conceptualization, Data curation, Investigation, Methodology, Project administration, Resources, Software, Validation, Writing - original draft, Writing - review & editing. **Brian Norton:** Supervision, Methodology, Writing - review & editing. **Amir Rahimi:** Supervision, Methodology, Writing - review & editing. **Hajir Karimi:** Supervision, Methodology, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

At Tyndall National Institute and Technological University Dublin,

this research was supported by MaREI, the SFI Research Centre for Energy, Climate and Marine [Grant No. 12/RC/2302_P2].

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